Week #7: [Mechanical Back & Neck Pain]

(Non-hyperlinked = Dr. Caitlin Cassidy Module)

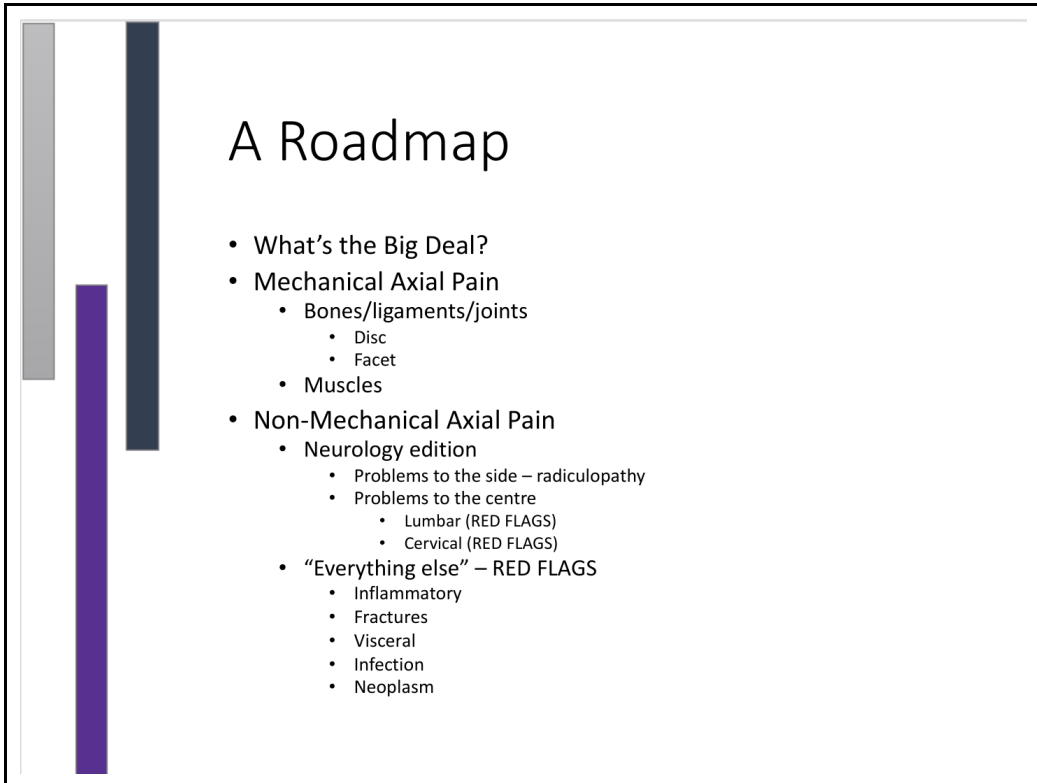
- Introduction to Module
- Axial Pain Overview (13 min)

Part I: So You've Got Mechanical Back Pain.... What's Going On Back There?

- **Anatomy of the Bones, Joints, & Ligaments of the Back (18 min)**
- Mechanical Spine Pain (12 min)

Part II: So You've got mechanical back pain... what's going on back there?

- **Anatomy of the Muscles of the Back (38 min)**
- Mechanical Spine Pain (12 min)



Why is this so important?

- Back and neck pain are very common presenting complaints in primary care
- Specific anatomic diagnosis can be tricky in “run of the mill” back pain (aka ”mechanical back pain”…more on that later)
  - Patients often get conflicting information
- Is a specific anatomic diagnosis truly “necessary” in uncomplicated back pain?
  - Maybe not! BUT you’d better be able to get a good idea of what’s going on to decide this…

Lots of Potential Pain Generators

Over here

Paraspinal musculature/soft tissues

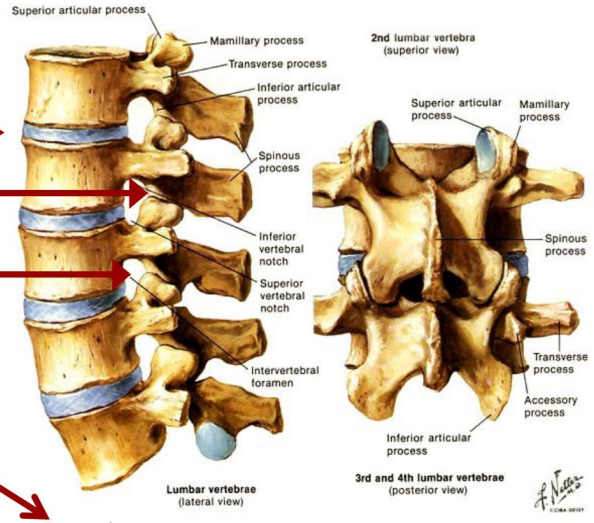
Disc

Facet (zygapophyseal) joint

Nervous structures

SI Joint

Down here somewhere



How Can We Sort Through This?

- Know your anatomy!!! (Ie. what's in there that could be causing the problem?)
- Patient history - a tool to focus our attention
  - O – onset
  - L – location
  - D – duration
  - C – character
  - A – alleviating/aggravating factors
  - R – radiation
  - S – severity
  - Red flags

Side Note… What Are Red Flags?

- Red flags for back and neck pain – raise alarm bells that a more sinister/less common cause of pain should be considered
  - Neurologic – progressive weakness or sensory change, saddle anaesthesia, loss of hand dexterity, ataxic gait, incontinence, erectile dysfunction, abnormal reflexes
  - Inflammatory – morning stiffness, pain in second half of night, family history, rashes, eye symptoms, improves with exercise
  - Neoplastic – young age, severe night pain and/or failure to improve with rest, unintentional weight loss, night sweats, history of cancer, associated signs (ie. hematuria)

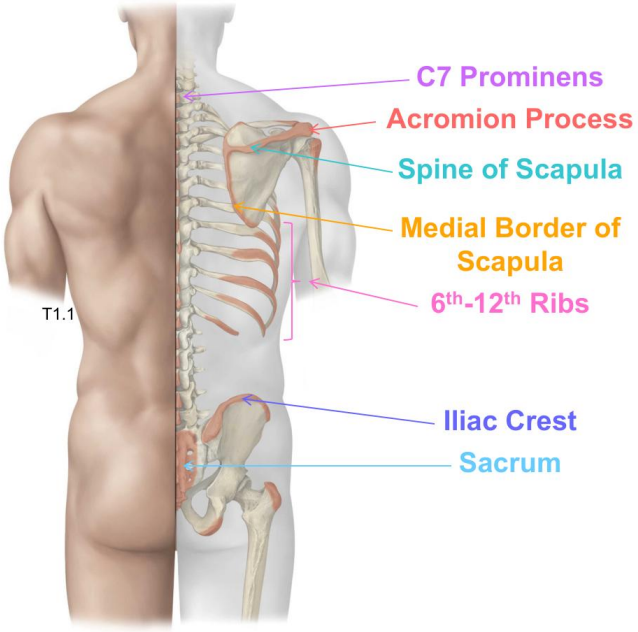
Side Note…
What Are Red Flags?

- Red flags for back and neck pain – raise alarm bells that a more sinister/less common cause of pain should be considered
  - Infectious – fevers, sweats, history of IV drug use, history of immunosuppression
  - Fracture (ie. subacute/chronic) – history of fall/trauma, history of osteoporosis, history of previous fragility fracture
  - Vascular
    - Vertebral artery insufficiency (cervical pain) – dizziness/lightheadedness with neck movement, cranial nerve abnormalities, drop attacks
    - Distal claudication – pain in extremities with activity (with or without associated neck/back pain)

How Can We Sort Through This?

- Physical Exam!
  - Looking for axial abnormalities and for contributory L/E abnormalities
  - Looking for neurologic abnormalities
  - Looking for signs of red flag conditions
- Clinical Skills!!!!

Palpable Back Structures



How Can We Sort Through This?

- Investigations
  - Mainly we're talking about spinal imaging
  - Other options?
    - In the presence of red flags, lots might be considered in addition to imaging
      - Infectious – blood work, cultures if applicable
      - Neoplastic – consider abdominal or other imaging, associated bloodwork or other testing
      - Vascular – may need specific vascular imaging (ie. angiography)
      - Neurologic – electromyography (nerve function tests)

Spinal Imaging

- What's available?
  - X-ray ($) – quick screening of alignment, check for fractures
  - CT ($$) – good for bony detail, trauma
  - MRI ($$$) – best detailed assessment of non-bony structures
  - Bone scan – assess for multifocal disease, hot vs cold lesions
  - CT Myelogram – typically only used if patient has contraindication for MRI (invasive dye study, injection into thecal sac)
  - Image guided injections for diagnostic clarity + therapeutic benefit

Spinal Imaging – What's indicated?

- Does everyone get imaged?
- Vast majority of cases of low back pain don't need imaging
  - Often can't identify the pathology (ie. soft tissue)
  - Can't distinguish symptomatic from asymptomatic lesions
  - Often doesn't change management
- It is REQUIRED though to look for bad things (RED FLAGS)

Mechanical Spine Pain

By definition, axial pain that is worse with movement

Road Map!

- What's the Big Deal
- Mechanical Axial Pain
  - Bones/ligaments/joints
    - Disc
    - Facet
  - Muscles
- Non-Mechanical Axial Pain
  - Neurology edition
    - Problems to the side – radiculopathy
    - Problems to the centre
      - Lumbar (RED FLAGS)
      - Cervical (RED FLAGS)
  - "Everything else" – RED FLAGS
    - Inflammatory
    - Fractures
    - Visceral
    - Infection
    - Neoplasm

Clinical Indicators that Neck/LBP is Mechanical

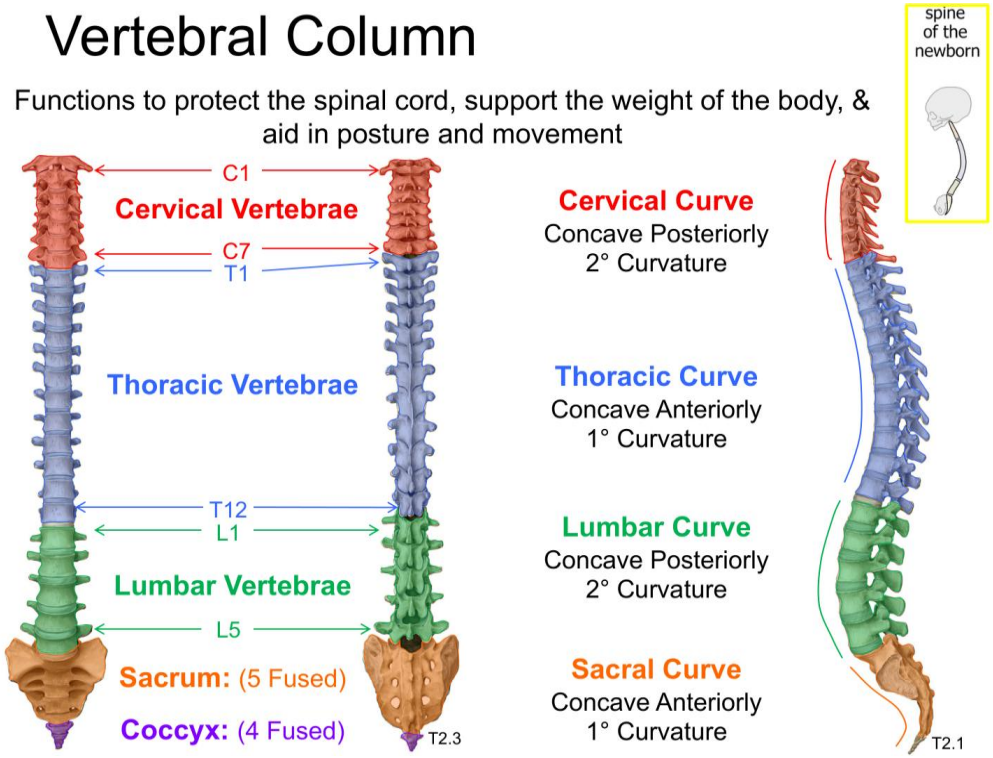
- O – anywhere on the acute to chronic spectrum
  - Often the result of an accumulation of changes over time (ie. degenerative processes)
  - Often "tweaked" by some acute irritating process like going down the stairs or bending wrong, etc
- L – midline or peri-midline neck or back
- D – variable
- C – dull, aching or sharp/stabbing
- A – aggravated by movement, relieved by rest
- R – variable
- S – variable
- Red flags – NONE

So You've Got Mechanical Back Pain…What's Going On Back There?

Part I: The Bones, Joints, & Ligaments

Vertebral Column

Functions to protect the spinal cord, support the weight of the body, & aid in posture and movement

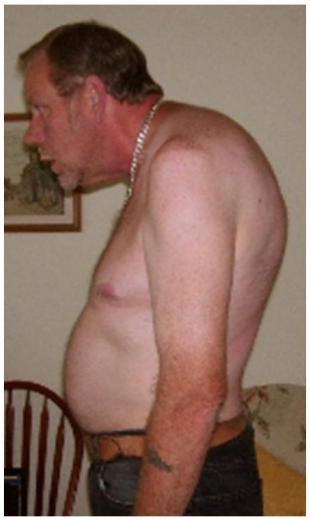


The vertebral column extends from the base of the skull to the coccyx and functions to protect the spinal cord, support the weight of the body, and aid in posture and movement

Consists of 33 vertebrae:

1. **7 Cervical** vertebrae
2. **12 Thoracic** vertebrae
3. **5 Lumbar** vertebrae
4. **5 Sacral** vertebrae – fused to form the **sacrum**
5. **4 Coccygeal** vertebrae – fused to form the **coccyx**

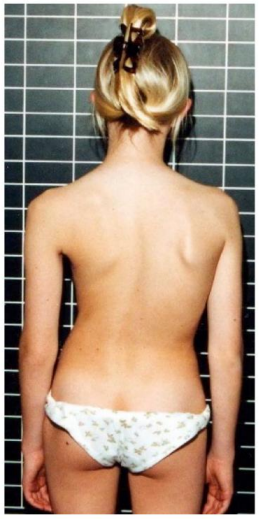
Abnormal Vertebral Curves



Kyphosis



Lordosis



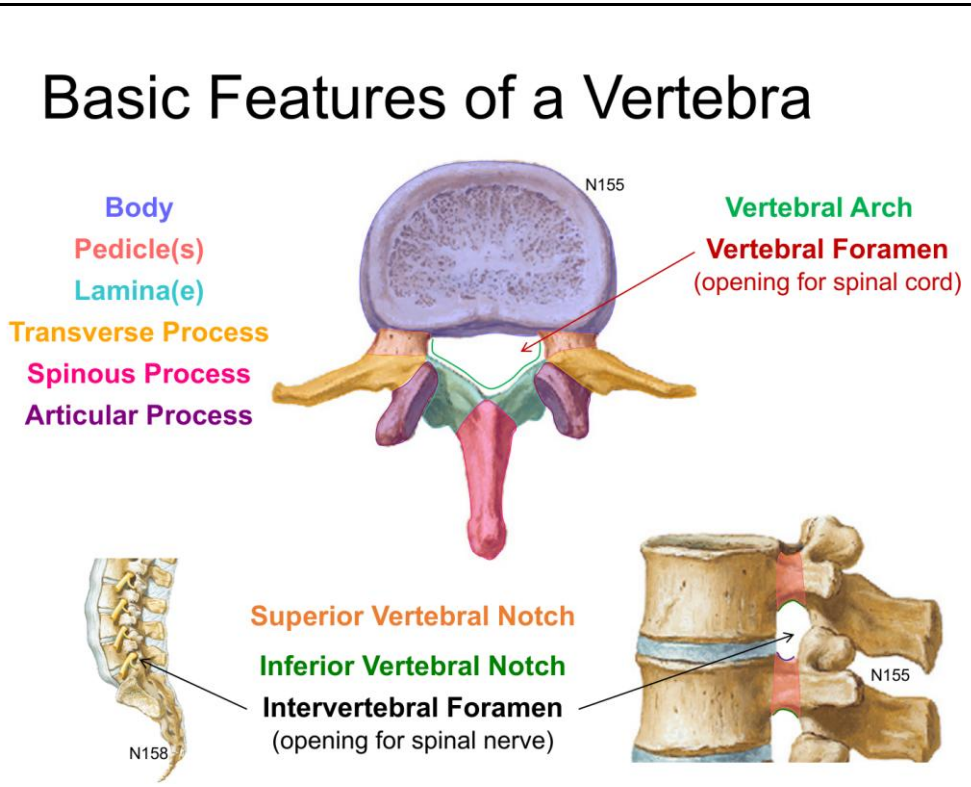
Scoliosis

Curvatures of the Vertebral Column:

1. Normal Curves:
   a. **Thoracic** and **sacral curvatures** are concave **anteriorly**
   b. **Cervical** and **lumbar curvatures** are concave **posteriorly**
2. Abnormal Curvatures:
   a. **Kyphosis** – exaggerated *thoracic curvature*, typically the result of erosion or fracture of the anterior part of the vertebrae, due to **osteoporosis** or other aging factors
   b. **Lordosis** – exaggerated *lumbar curvature*, characterized by anterior rotation of the pelvis. Often cause by weakened trunk musculature and seen during pregnancy
   c. **Scoliosis** – increased *lateral curvature* of the spine with rotation of the vertebrae. May be due to weakness of back muscles, poor posture, congenital defects in vertebrae development, and inequality in the length of the lower limbs

Features of a Vertebra:

1. **Vertebral Body** – large anterior portion that provides strength and permits transmission of body weight. Vertebral bodies become larger as they progress inferiorly in order to support increasingly greater body weight
2. **Vertebral Arch** – posterior portion of vertebra, composed of fused pedicles

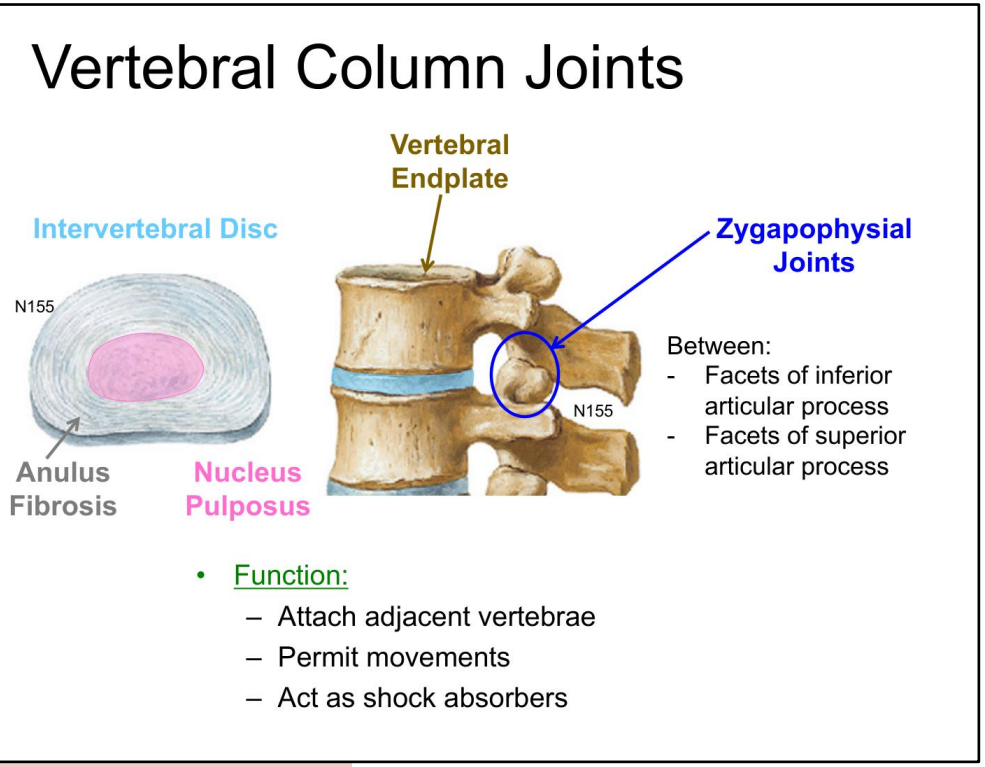


and lamina

   a. ***Pedicles*** – short, stout processes that join the vertebral arch to the vertebral body. Each pedicle has superior and inferior vertebral notches that form the intervertebral foramen
   b. ***Lamina*** – 2 broad, flat plates that close the vertebral arch posteriorly

3. **Vertebral Foramen** – opening for spinal cord; surrounded by the posterior wall of the vertebral body and the vertebral arch
4. **Spinous Process** – projects posteriorly from the vertebral arch at the junction of the two laminae; attachment site for muscles and ligaments
5. **Transverse Processes** – two processes that project posterolaterally from junction of pedicles and laminae; attachment site for muscles and ligaments
6. **Articular processes** – four processes, 2 superior and 2 inferior, that articulate with adjacent vertebrae forming zygapophysial joints. Located at the junction of the pedicles and laminae

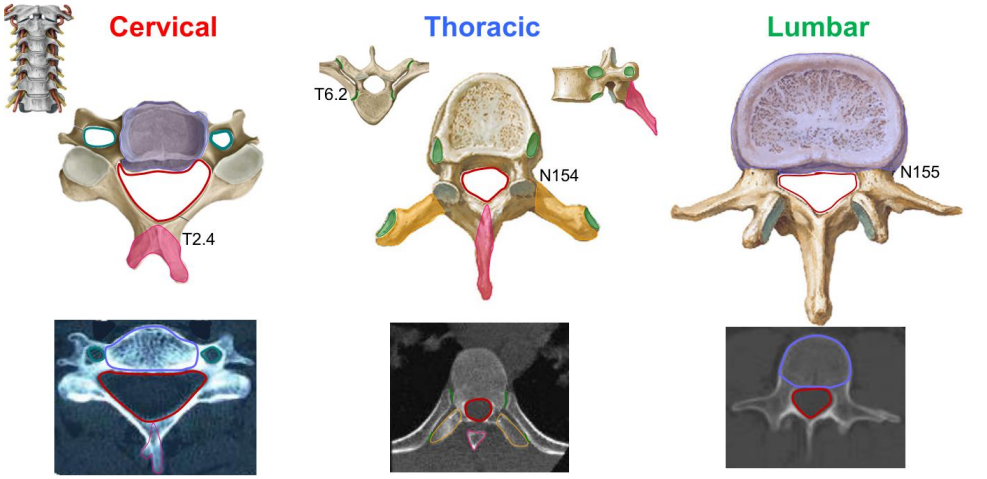
Clinical Note: a **laminectomy** is the surgical excision of one or more spinous processes and their supporting laminae in a particular region of the vertebral column. Laminectomies provide access to the vertebral canal to relieve pressure on the spinal cord or nerve roots, commonly caused by a tumor, herniated IV disc, or boy hypertrophy (excess growth).



Joints of the Vertebral Column:

1. Each vertebra (except C1 that attaches to the skull and L5 that attaches to the sacrum) attaches to the superiorly and inferiorly adjacent vertebra at the bodies of the vertebrae via **intervertebral discs** and at the **zygapophysial joints**. The intervertebral discs interface with the vertebral bodies at a transitional area known as the vertebral end plate – The vertebral end plate is comprised of two layers a cartilaginous layer that fuses with the disc and a thin layer of porous bone attached to the vertebral body.
2. **Intervertebral (IV) Discs** are located between the vertebral bodies. They are strong attachments between the vertebrae, they permit movement, and serve as shock absorbers
    a. **Components:**
        i. **Anulus Fibrosis** – outer portion, consisting of concentric layers of fibrocartilage
        ii. **Nucleus Pulposus** – inner portion, consisting of a gelatinous matrix with high water content
    a. **Components (continued)**
        a. Thickness of IV discs varies with vertebral column region (cervical and lumbar regions are thicker) iii. IV discs account for ~25% of the vertebral column height
        b. There is no IV disc between C1 and C2
        c. **Clinical Note: Disc herniation**, or protrusion of the nucleus pulposus into or through the annulus fibrosus, is a well-recognized cause of back pain. Herniated material may directly compress nerve roots or the spinal cord itself, resulting in pain and/or neurological deficits. Most posterolateral herniations occur between L4//L5 or L5/S1, typically affecting the nerve root immediately inferior to it (i.e., the L4 disc impinges on the L5 nerve root).
2. **Zygapophysial Joints** are plane synovial joints between the superior and inferior articular processes of adjacent vertebrae.
3. **Craniovertebral joints:**
    a. **Atlanto-occipital joints** – formed between the atlas (C1) and the occipital condyles
        i. Permit flexion and extension (nodding) of the head
    b. **Atlanto-axial joints** – formed between the atlas (C1) and axis (C2)
        i. Permit the head to turn from side to side (saying 'no') via pivoting of C1 around the dens of C2
    c. **Costovertebral joints** – a typical rib articulates posteriorly with the vertebral column at two joints:
        i. Joints at the heads of the ribs:
            1. The head of the rib articulates with the superior costal facet of the corresponding (same- numbered) vertebra
            2. The inferior costal facet of the vertebra superior to it
            3. The adjacent IV disc uniting the two vertebrae
    d. **Costotransverse joints** - the tubercle of the rib articulates with the articular facet on the transverse process of the corresponding (same-numbered) rib

Characteristics of Regional Vertebrae

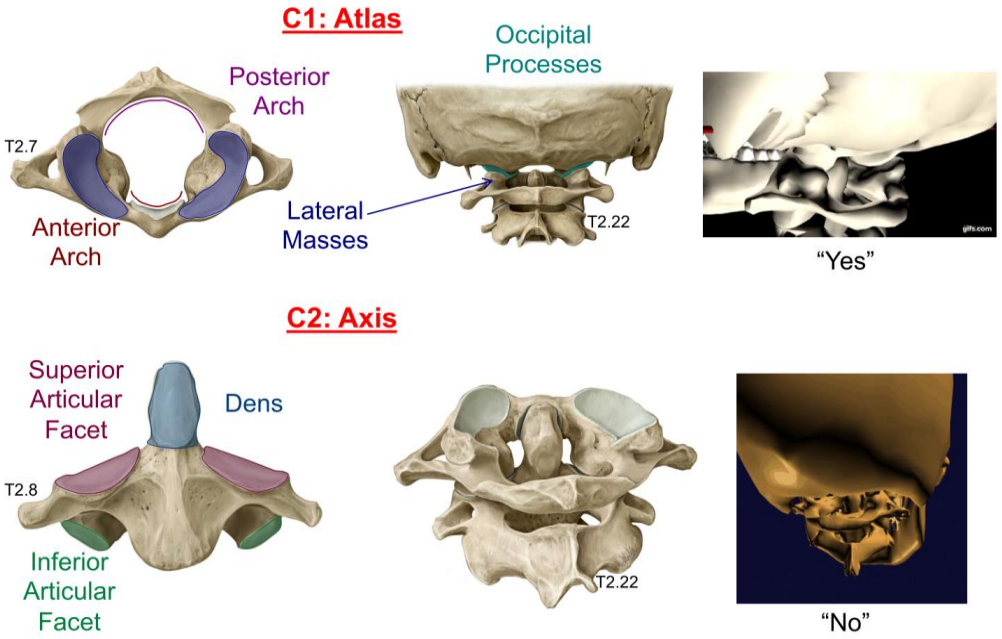


- Small bodies
- Large vertebral foramen
- Transverse foramina
- Bifid spinous process

- Small vertebral foramen
- Long transverse process
- Long spinous process
- Costal facets (articulate with ribs)

- Large bodies
- Large vertebral foramen
- No costal facets

Atypical Vertebrae: C1 & C2



Regional Vertebrae Differences:

1. Cervical:

a. ***Vertebral body:*** small
b. ***Vertebral foramen:*** large; especially C3-C7 due to the enlargement of the spinal cord
c. ***Transverse processes:*** short, with transverse foramina (openings) in all cervical vertebrae
   i. Vertebral artery ascends to the skull through the transverse foramen from C6-C1
   ii. Only the vertebral vein passes through the C7 transverse foramina
d. ***Articular processes:*** superior articular facets directed superoposteriorly; inferior facets directed inferoanteriorly
e. ***Spinous processes:*** short and bifid, with exception of C7, which has an exaggerated spinous process called the vertebra prominens (easily palpated when neck is flexed)
f. ***Movements:*** flexion and extension
g. C1 and C2 are atypical vertebrae:
   i. Atlas (C1) – ring shaped; consisting of two **lateral masses** (with superior and inferior articular processes) that are connected by an **anterior arch** (replaces body) and a **posterior arch** (instead of lamina). On its cranial surface is a groove for the vertebral artery
   ii. Axis (C2) – its distinguishing feature is the **dens**, which extends superiorly from the C2 body and forms a pivot joint with the articular facet on the posterior surface of the anterior arch of C1

2. Thoracic:

a. ***Vertebral body:*** heart-shaped with **costal facets** for articulation with the ribs via **costovertebral joints** and **costotransverse joints**
b. ***Vertebral foramen:*** circular and smaller than those in the cervical and lumbar regions
c. ***Transverse processes:*** long; transverse costal facets for articulation with the tubercle of rib
d. ***Articular processes:*** superior articular facets are directed posteriorly and slightly laterally; inferior articular facets are directed anteriorly and slightly medially.

This orientation prevents excessive flexion and extension

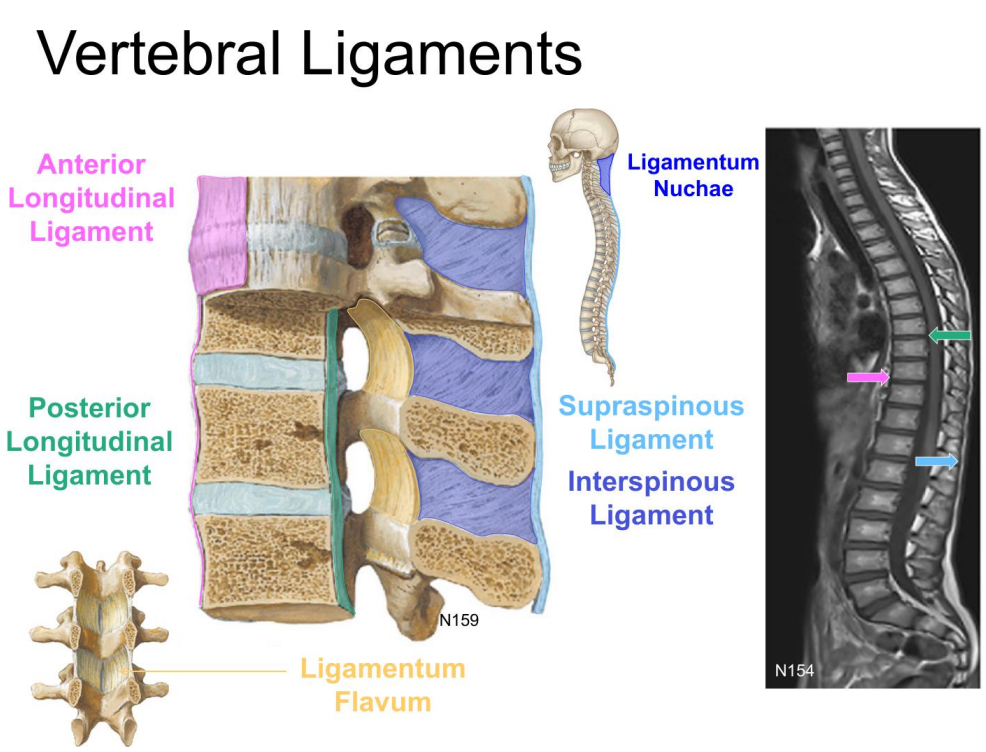
e. ***Spinous processes:*** long; slopes inferiorly and overlaps vertebral arch inferior to it
f. ***Movements:*** rotation and some lateral flexion

3. Lumbar:

a. ***Vertebral body:*** larger, strong vertebrae with short, heavy pedicles
b. ***Vertebral foramen:*** triangular-shaped (larger than thoracic vertebral foramina)
c. ***Transverse processes:*** long and slender transverse processes, with no costal facets
d. ***Articular processes:*** superior articular facets are directed posteromedially; inferior articular facets are directed anterolaterally
e. ***Spinous processes:*** short and stout
f. ***Movements:*** flexion, extension, some lateral bending, and limited rotation

G. Sacrum:

a. Consists of 5 fused vertebrae that provide strength and stability to the pelvis and transmits body weight to the ilium through the **sacroiliac joints**
b. Anterior (pelvic) surface is smooth and concave, while the posterior surface is rough and convex
c. **Sacral Foramina** – paired openings on the anterior and posterior sides to transmit the sacral nerves
d. **Sacral Hiatus** – inverted U-shaped opening that leads into the sacral canal (inferior end of the vertebral canal); this is one side for a caudal epidural anesthesia



Vertebral Column Ligaments:

a. **Anterior Longitudinal Ligament** – strong, broad fibrous band that covers the anterolateral aspect of the vertebral bodies and IV discs
    i. Runs from the occipital bone to the sacrum
    ii. Helps resist *hyperextension* of the vertebral column
b. **Posterior Longitudinal Ligament** – narrow ligament that runs within the vertebral canal along the posterior aspect of the vertebral bodies
    i. Helps resist hyperflexion of the vertebral column
c. **Ligamentum Flava** – broad, pale yellow fibrous tissue joining adjacent vertebral laminae
d. **Supraspinous Ligaments** – connects the tips of the spinous processes from C7 to the sacrum
    i. Above C7, these fibres merge with the ***nuchal ligament***, which is an extremely strong band stretching from the external occipital protuberance to the spinous processes of the cervical vertebrae; the ***nuchal ligament*** extends from the spinous processes to the skin filling the cervical curvature; serves as an attachment point for the muscles of the neck and upper back
e. **Interspinous Ligaments** – thin ligament that connects adjacent spinous processes
f. **Intertransverse Ligaments** – connect adjacent transverse processes

Knowledge Check

A





Large vertebral body
Lumbar vertebra

B

Cervical vertebra



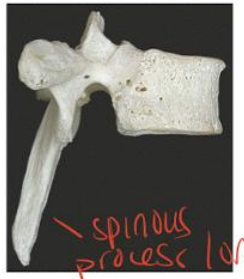


Small vertebral body
Transverse process short & spinous process

C

Thoracic





Knowledge Check

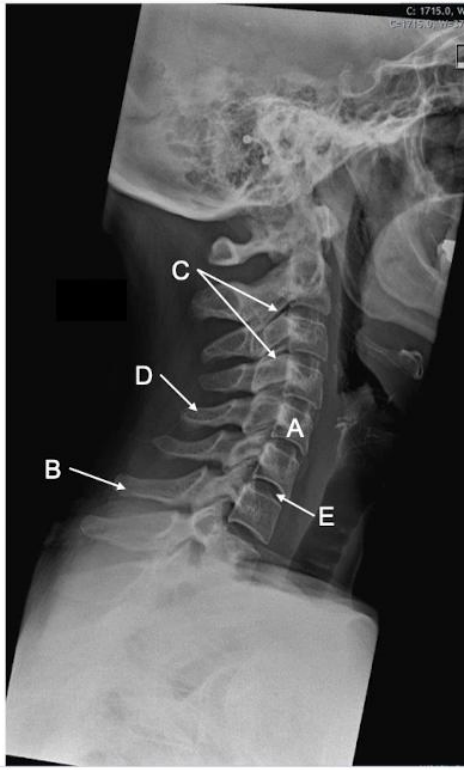
C7 Prominens (B) C7 prom

Facet Joints (C) - facet joints b/n vertebra

Intervertebral Disc (E)

Spinous Process (D) - sticks out a lot

Vertebral Body (A) of C5



Knowledge Check

- more space on lumbar vertebra

Facet Joints (D)

Inferior Vertebral Notch (C) - inferior asp. of pedicle

Intervertebral Disc (G) - larger space

Intervertebral Foramen

Sacrum (H) - 1-5 + sacrum promontory

Spinous Process (A) thicc up

Superior Vertebral Notch (E)

Vertebral Body (B) - sig larger



Mechanical Spine Pain

By definition, axial pain that is worse with movement

Road Map!

- What's the Big Deal
- Mechanical Axial Pain
  - Bones/ligaments/joints
    - Disc
    - Facet
  - Muscles
- Non-Mechanical Axial Pain
  - Neurology edition
    - Problems to the side – radiculopathy
    - Problems to the centre
      - Lumbar (RED FLAGS)
      - Cervical (RED FLAGS)
  - "Everything else" – RED FLAGS
    - Inflammatory
    - Fractures
    - Visceral
    - Infection
    - Neoplasm

Repetitive axial loading of spine

↓

Damage to vertebral end plates

↓

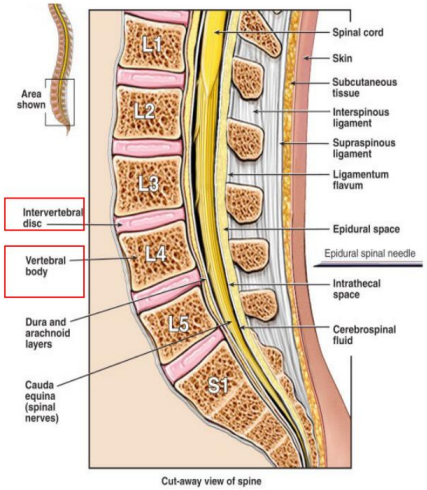
Microtears in annulus fibrosus

↓

Accumulation of 'weakness' in annulus→internal herniation of internal nucleus pulposus (IDD)

↓

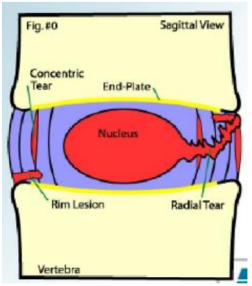
Loss of disc height as gelatinous nucleus becomes more fibrous and herniated to various degrees



Cut-away view of spine

Discogenic Mechanical Spine Pain

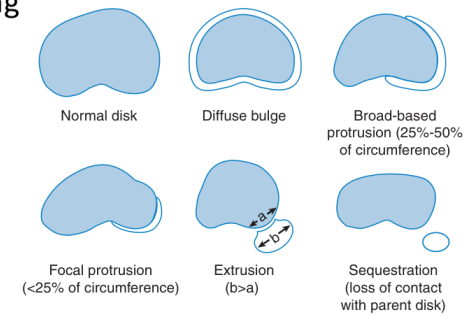
- Disc as a pain progenitor (ie. pain from the disc itself NOT from it squishing something else)
  - Internal Disc Disruption (annulus intact; nucleus pulposus pushes out through radial or concentric tears)
  - May account for back pain in 20-40% of patients
  - Classically described clinical features:
    - Axial predominant pain (ie. back>leg)
    - Mechanical (movement related)
      - Aggravated by forward flexion
  - Management is typically conservative
    - Therapy, exercise
    - NSAIDs acutely, acetaminophen



CFEP Back Tool, 2016

Disc Herniation

- Can cause BOTH mechanical pain (discogenic) AND compression of nervous structures
- Acute herniation more likely in younger population
  - Lifting something, sudden onset of axial pain
  - Worse with coughing/sneezing
- Commonly herniate posterolaterally in lumbar spine (away from ligamentum flavum, unfortunately towards nerve roots)



Braddom 2011

Repetitive axial loading of spine

↓

Damage to vertebral end plates

↓

Microtears in annulus fibrosus

↓

Accumulation of 'weakness' in annulus→internal herniation of internal nucleus pulposus (IDD)

↓

Loss of disc height as gelatinous nucleus becomes more fibrous and herniated to various degrees

Discs aren't in there by themselves…changes at the disc must affect surrounding structures…

Change in angles/repetitive microtrauma

↓

Synovial hypertrophy

↓

Cartilage destruction

↓

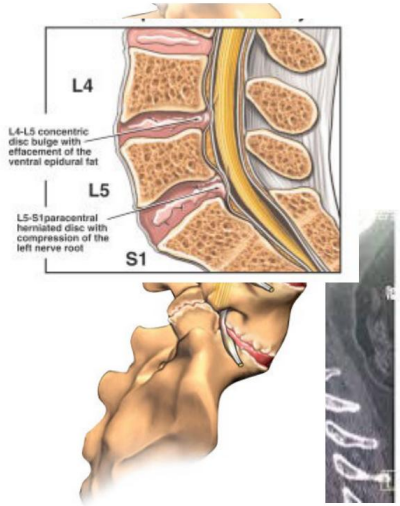
Osteophyte formation

↓

Capsular laxity +/- subluxation

↓

Enlargement of articular processes

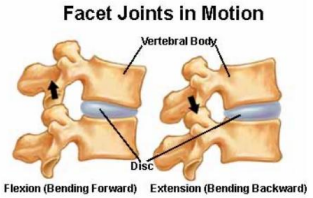


Which Brings Us To Spondylosis

- Spondylosis
  - "Spondy" = "vertebrae"
  - "Osis" = "a state of disease"
  - Spondylosis = refers to a combination of degenerative changes in the spine; encompasses "degenerative disc disease" + facet arthropathy with associated periarticular changes (loss of disc height, ligamentous hypertrophy)

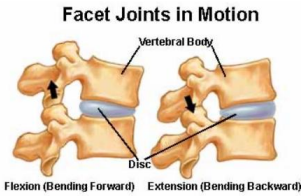
Facetogenic Mechanical Spine Pain

- Pain coming from the facet joint
  - Facets are synovial joints prone to degeneration (arthritis)
  - Findings of facet joint degeneration not necessarily correlated with clinical symptoms (Kalichman et al 2008)
    - Hence why imaging may not be that helpful!
  - Clinical picture
    - Localized axial (?unilateral) pain +/- some radiation to limbs
      - Commonly results in headache in cervical spine
      - Uncommon to radiate past knee in lumbar spine
    - Classically worse w/ ext, relieved by flexion



Facetogenic Mechanical Spine Pain

- Pain coming from the facet joint
  - Facets are synovial joints prone to degeneration (arthritis)
  - Findings of facet joint degeneration not necessarily correlated with clinical symptoms (Kalichman et al 2008)
    - Hence why imaging may not be that helpful!
  - Clinical picture
    - Localized axial (?unilateral) pain +/- some radiation to limbs
      - Commonly results in headache in cervical spine
      - Uncommon to radiate past knee in lumbar spine
    - Classically worse w/ ext, relieved by flexion
  - Management
    - Therapies (flexion based)
    - NSAIDs acutely, acetaminophen
    - ?joint injection, medial branch block/ablation



So You've Got Mechanical Back Pain…What's Going On Back There?

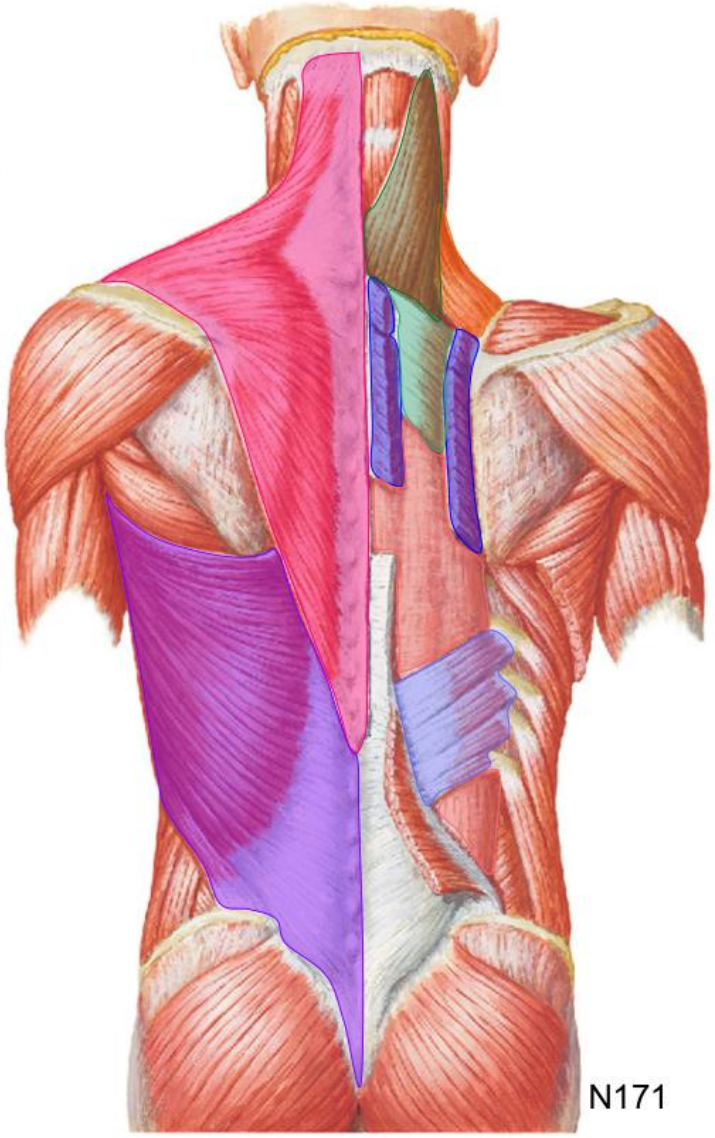
Part II: The Muscles

Anatomy of the Muscles of the Back

Back Musculature

Superficial Musculature

- Trapezius
- Latissimus Dorsi
- Rhomboids
- Levator Scapulae



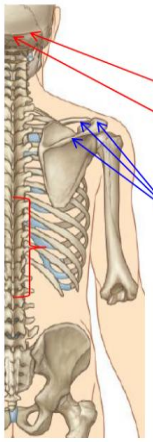
Deep Musculature

- Splenius
- Erector Spinae
- Transversospinalis

Intermediate Musculature

- Serratus Posterior Superior
- Serratus Posterior Inferior

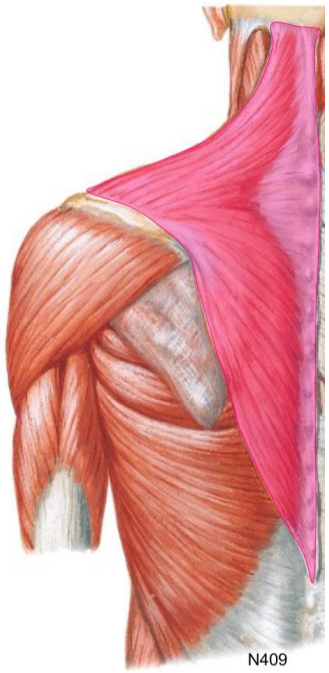
Trapezius



Origin:
Superior Nuchal Line
External Occipital Protuberance
Ligamentum Nuchae
Spinous Processes C7-T12

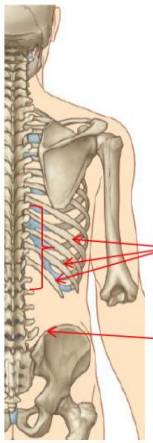
Insertion:
Lateral ⅓ of Clavicle
Acromion Process
Spine of Scapula

- Function:
  - Elevates, depresses and retracts scapula
  - Rotates glenoid cavity superiorly
- Innervation:
  - Spinal Accessory Nerve (CN XI)

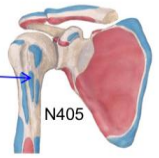


N409

Latissimus Dorsi



Insertion:
Intertubercular Groove

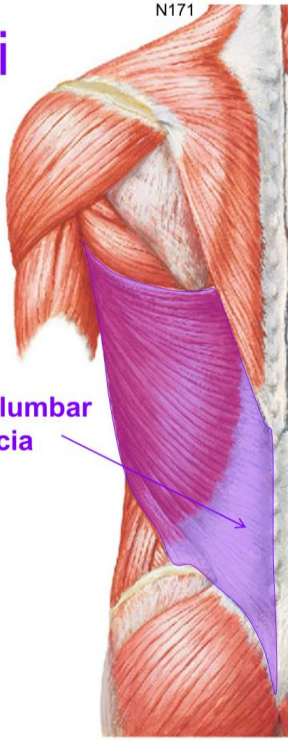


N405

Origin:
Inferior 3-4 Ribs
Spinous Processes of T7-12
Thoracolumbar Fascia
Iliac Crest

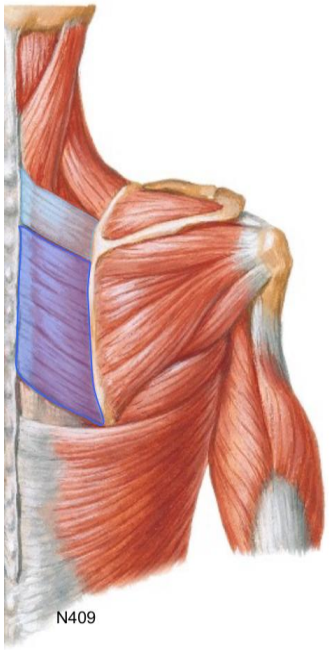
- Function:
  - Extends, adducts, and medially rotates humerus
- Innervation:
  - Thoracodorsal Nerve

N171



Thoracolumbar Fascia

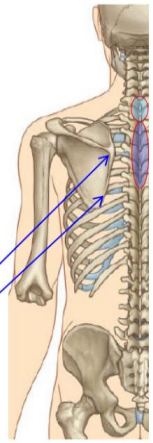
Rhomboids



N409

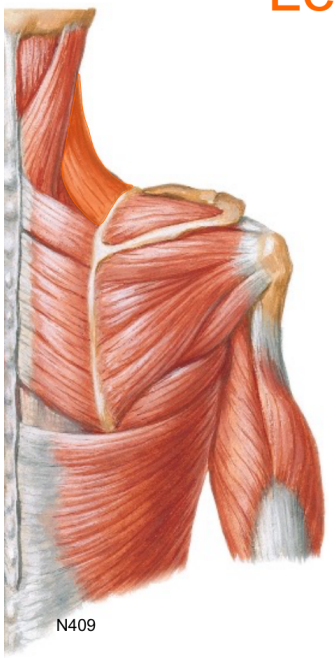
Origin:
Minor: Ligamentum Nuchae
C7-T1 Spinous Processes
Major: T2-5 Spinous Processes

Insertion:
Minor: Medial border at root of the spine
Major: Medial border inferior to root of the spine



- Function:
  - Retract scapula
  - Rotate to depress glenoid cavity
  - Fix Scapula to thoracic wall
- Innervation:
  - Dorsal Scapular Nerve

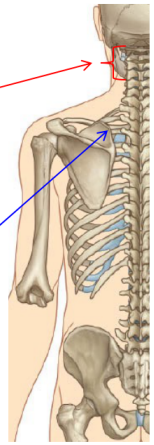
Levator Scapulae



N409

Origin:
C1-4 Transverse Processes

Insertion:
Medial border superior to root of the spine



- Function:
  - Elevates scapulae
  - Rotates to depress glenoid cavity
- Innervation:
  - Dorsal Scapular Nerve
  - C3-4 Spinal Nerves

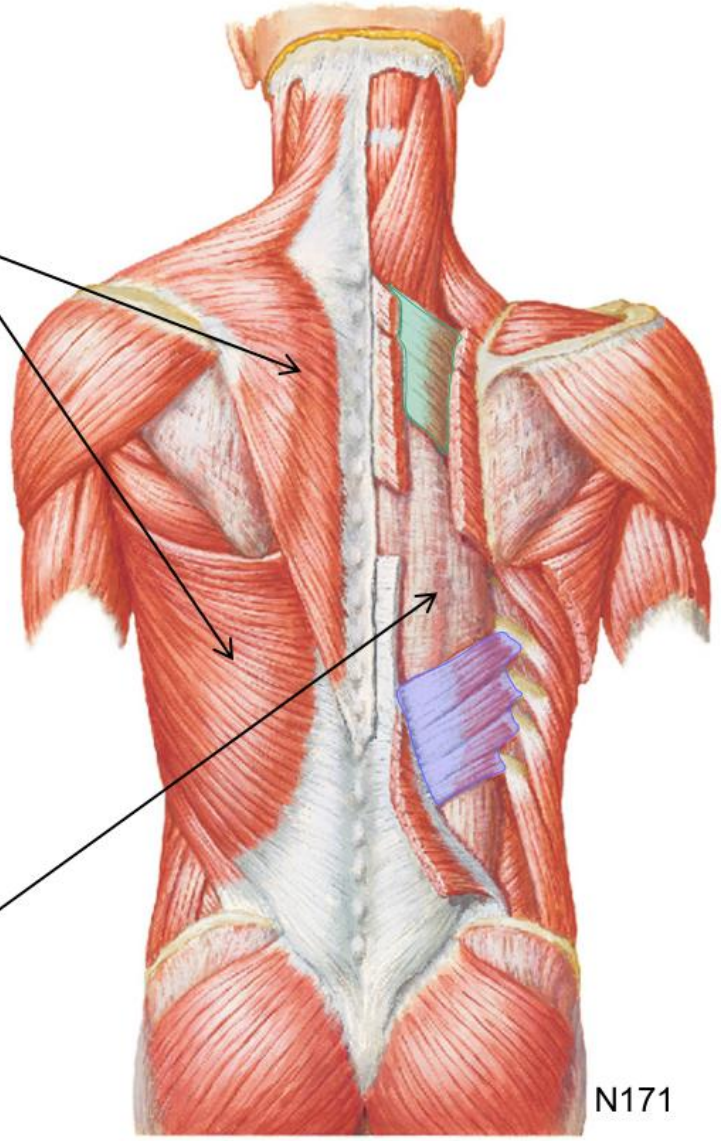
Superficial (Extrinsic) Muscles of the Back

Muscle	Description	Main Action(s)	Innervation
Trapezius	Large trapezoidal-shaped muscle; covers neck and upper trunk	Elevates, depresses, and retracts scapula; rotates glenoid cavity superiorly	Spinal Accessory Nerve
Latissimus Dorsi	Large, flat, fan-shaped muscle; Origin is dense and aponeurotic	Extends, adducts, and medially rotates humerus	Thoracodorsal Nerve
Levator Scapulae	Relatively thin, strap like muscle	Elevates scapula and rotates to depress glenoid cavity	Dorsal Scapular Nerve & C3-4 Spinal Nerves
Rhomboids	Deep to trapezius; two muscle bellies (major & minor) often fused	Retract scapula and rotate to depress glenoid cavity; fix scapula to thoracic wall	Dorsal Scapular Nerve

Intermediate Musculature

Superficial Musculature

Deep Musculature



Intermediate Musculature

Serratus Posterior Superior

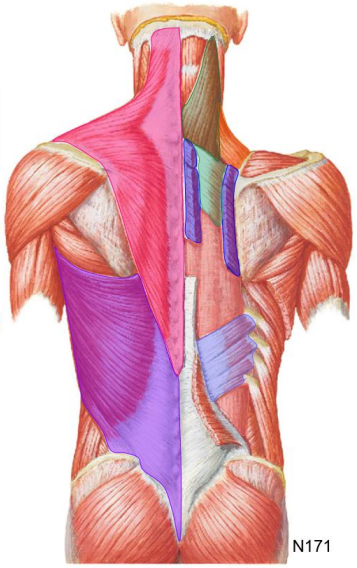
Serratus Posterior inferior

Intermediate (Extrinsic) Muscles of the Back

Deep Back Musculature

Superficial Musculature

- Trapezius
- Latissimus Dorsi
- Rhomboids
- Levator Scapulae



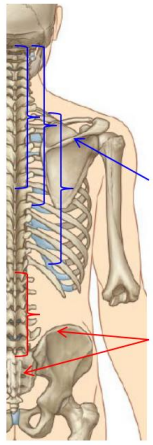
Deep Musculature

- Splenius
- Erector Spinae
- Transversospinalis

Intermediate Musculature

- Serratus Posterior Superior
- Serratus Posterior Inferior

Erector Spinae

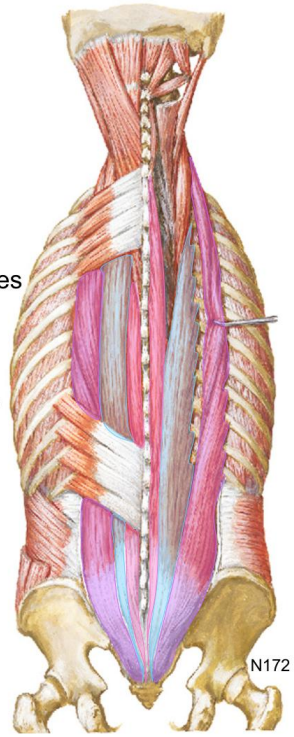


- Iliocostalis
- Longissimus
- Spinalis

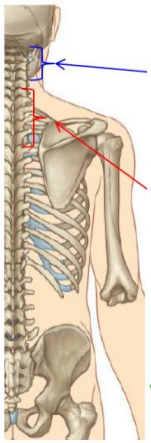
Insertion:
Iliocostalis: rib & superior transverse processes
Longissimus: ribs & superior transverse processes
Spinalis: superior spinous processes

Origin:
Broad tendon from sacrum, iliac crests, & inferior spinous processes

- Function:
  - Acting Bilaterally: Extend vertebral column & head
    Eccentrically control trunk flexion
  - Acting Unilaterally: Laterally flex vertebral column
- Innervation:
  - Dorsal Rami of Spinal Nerves

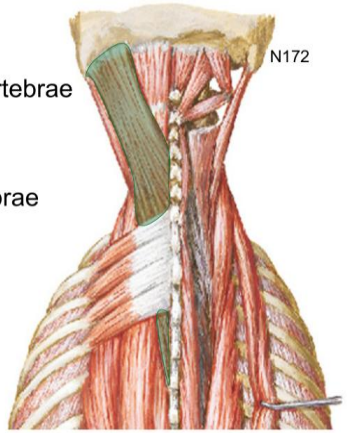


Splenius Muscle Group

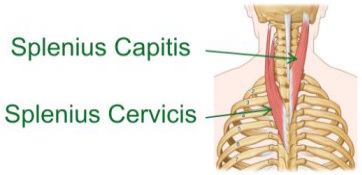


Insertion:
Occipital Bone
Transverse Processes of Upper Cervical Vertebrae

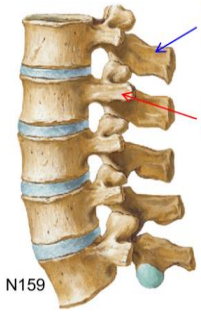
Origin:
Ligamentum Nuchae
Spinous Processes of Upper Cervical Vertebrae



- Function:
  - Acting Bilaterally: Extend head & neck
  - Acting Unilaterally: Laterally flex & rotate head & neck
- Innervation:
  - Dorsal Rami of Spinal Nerves



Transversospinalis Muscle Group



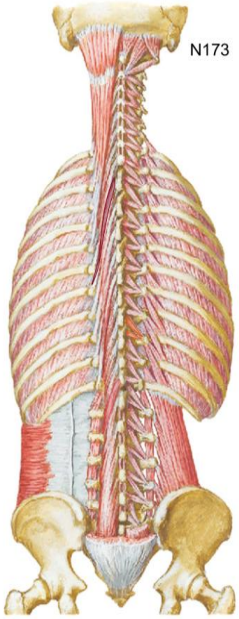
Insertion:
Spinous Process

Origin:
Transverse Process

Semispinalis
Spans 4-6 segments

Multifidis
Spans 2-4 segments

Rotatores
Spans 1-2 segments



- Function:
  - Contralateral rotation
  - Stabilization
  - Proprioception
- Innervation:
  - Dorsal Rami of Spinal Nerves

Deep (Intrinsic) Muscles of the Back

1. Maintain posture and control movements of the vertebral column
2. Innervated by posterior rami of spinal nerves; these posterior rami ultimately become cutaneous nerves that provide sensation to the skin of the back and neck.

provide sensation to the skin of the back and neck

Muscle	Description	Main Action(s)	Innervation
Erector Spinae	Three muscles that function as a group **Iliocostalis:** Most lateral **Longissimus:** intermediate muscle; usually widest **Spinalis:** most medial	*Acting bilaterally*: extend vertebral column and head; eccentrically control trunk flexion *Acting unilaterally*: laterally flex vertebral column	Dorsal Rami of Spinal Nerves
Splenius	Composed of two parts: **cervicis** and **capitis**; both ascend in a medial to lateral direction	*Acting bilaterally*: extends head and neck *Acting unilaterally*: laterally flexes and rotates head and neck	Dorsal Rami of Spinal Nerves
Transversospinalis	Three muscles of different lengths are a part of this muscle group: Semispinalis, Multifidus, Rotatores; span 1-6 vertebral levels	Contralateral rotation and stabilize vertebrae and proprioception	Dorsal Rami of Spinal Nerves

Knowledge Check

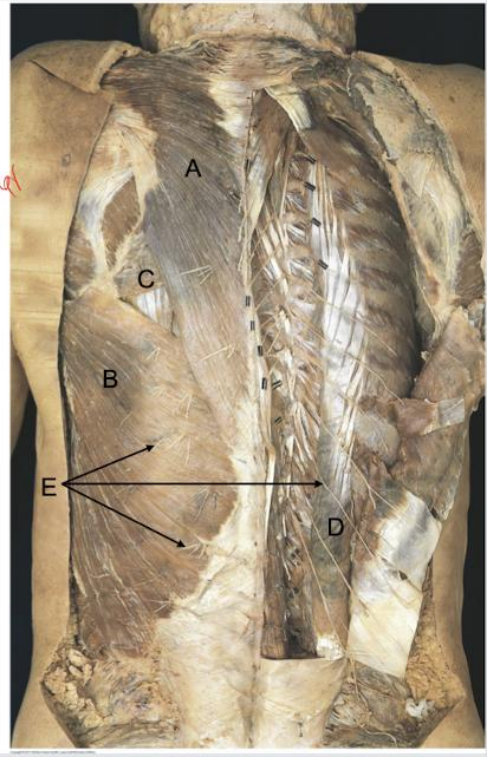
Erector Spinae (D) – extend / contract as you do eg. controlled flexion of trunk

Dorsal Rami (E)

Latissimus Dorsi (B)

Trapezius (A) 3 fibers: inferior, middle, superior

Rhomboids (C)



Mechanical Spine Pain

By definition, axial pain that is worse with movement

Road Map!

- What's the Big Deal
- Mechanical Axial Pain
  - Bones/ligaments/joints
    - Disc
    - Facet
  - Muscles
- Non-Mechanical Axial Pain
  - Neurology edition
    - Problems to the side – radiculopathy
    - Problems to the centre
      - Lumbar (RED FLAGS)
      - Cervical (RED FLAGS)
  - "Everything else" – RED FLAGS
    - Inflammatory
    - Fractures
    - Visceral
    - Infection
    - Neoplasm

Muscle Problems

- Acute strain/spasm
- Repetitive strain injuries
  - (Per WSIB): related to repetitive tasks, forceful exertions, vibrations, mechanical compression, sustained/awkward positions
- Myofascial pain related to trigger points
  - ”Hyperirritable areas of tissue that are tender when compressed and can give rise to referred pain” (Scott et al)

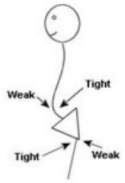
Scott, A, et al. “**Trigger Point Injections for Chronic Non-Malignant Musculoskeletal Pain: A Systematic Review”, PAIN MEDICINE,** 10(1):2009.

What To Do About Muscle Problems?

- Investigations?
  - Only if indicated to rule other things OUT
  - There are no “tests” for diagnosing these types of muscle problems

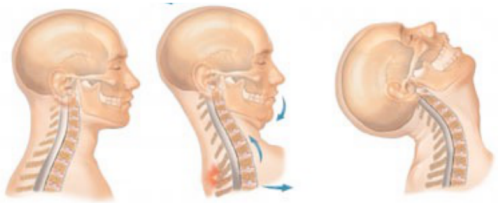
What To Do About Muscle Problems?

- Conservative management
  - Rule out red flags; if any concern, consider appropriate imaging
  - Education
    - Reassure
    - Discuss excellent prognosis
  - Relative rest
  - Modalities (ice, heat)
  - Topicals
  - Short course oral anti-inflammatories if necessary for acute pain
  - Physiotherapy
    - Strengthen postural support



What About Whiplash?

- "Whiplash Associated Disorder:" term given to collection of symptoms affecting neck; triggered by an accident with an acceleration/deceleration mechanism such as MVC
- Tissues affected
  - Most common – muscles (strains or partial tears)
  - Less common – facets > ALL > bones (fractures) > cord (SCI)



Pastakia et al, 2011

Whiplash Associated Disorder

- Majority improve with conservative strategies (<5% with severe ongoing symptoms at 2 years)
- Management of WAD
  - If it's more than subjective pain, stiffness/tenderness, reasonably reduced ROM, consider XR (if concerned about fracture) +/- MRI (if any neuro abnormalities)
  - If it is "straightforward"
    - Simple mobilization – keep moving and avoid C-collars
    - Modalities (heat, ice)
    - Meds as needed – tylenol, NSAID (oral and/or topical)
    - "Modest" physiotherapy – aggressive programs may worsen symptoms

The Mechanical Pain Conundrum

- Mechanical pain occurs due to irritation of one or more structures in the back
  - Worse with movement, better with rest
  - There is often NO ABNORMALITY on imaging
    - Endean et al 2011 *Spine* – review of prevalence of spine abnormalities reported on imaging (MR) among adults with NO BACK PAIN
      - Disc degeneration present in 0-76%
- Degenerative disc disease is a process that occurs with aging and may be accelerated by injuries or other processes
  - There are often NO SYMPTOMS present despite abnormal imaging
  - That said, it MAY be contributing to mechanical back pain

So How Should I Decide Who to Image?

- If there are NO RED FLAGS and the mechanical symptoms have been present for <6 weeks, NO IMAGING IS INDICATED
- Per the Canadian Association of Radiologists:

Lumbar spine				
C06. **Lower back pain** **(For children see L17)**	MRI	Indicated in special circumstances [B]	0	If imaging is indicated, MRI is the best modality. Imaging is only indicated if there are "red flag" indications: If there is clinical concern about an epidural abscess or hematoma which may present with acute pain but no neurological symptoms, urgent imaging is required. • Suspected cancer • Suspected infection. • Cauda equina syndrome • Severe/progressive neurologic deficit • Suspected compression fracture In patients with suspected uncomplicated herniated disc or spinal stenosis imaging is only indicated after an unsuccessful 4- 6 week trial of conservative management.

https://car.ca/wp-content/uploads/Spine.pdf

Mechanical Pain

So How Should I Decide Who to Image?

- If there are NO RED FLAGS and the mechanical symptoms have been present for <6 weeks, NO IMAGING IS INDICATED
- Per the Canadian Association of Radiologists:
  - CT is only indicated when you would do an MRI but can't get one due to contraindications or lack of availability
  - XR is only indicated if you're looking for red flag conditions (fractures, malignancy, etc)
    - But may be an appropriate starting point for someone with symptoms persisting >6 weeks DESPITE good clinical/conservative treatment

https://car.ca/wp-content/uploads/Spine.pdf

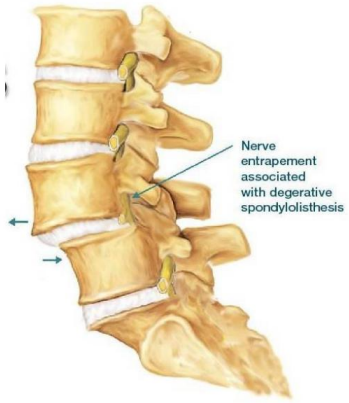
Spondylosis with degenerative changes in the disc and/or in the facets

↓

Subluxation, expansion of joint (osteophytosis, ligament hypertrophy), disc bulge/herniation, degenerative spondylolisthesis

Degenerative Spondylolisthesis

- Spondy = vertebrae
- Listhesis = dislocation, slipping







Spondylosis with degenerative changes in the disc and/or in the facets

↓

Subluxation, expansion of joint (osteophytosis, ligament hypertrophy), disc bulge/herniation, degenerative spondylolisthesis

↙ ↘

Nerve root encroachment in neural foramina (lateral stenosis)

Central canal stenosis